

6.4 Least squares via projection

We now return to the linear model $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, with $\mathbf{X}$ an $n \times p$ matrix of rank $r \leq p$, and $\mathbf{M}$ the orthogonal projection onto $\mathcal{C}(\mathbf{X})$. Nothing in this section uses probability. The results hold for every data vector $\mathbf{y}$ and so apply whatever the distribution of the errors.

6.4.1 The characterization of least squares estimates

Theorem 6.4.1. Least squares is projection

A vector $\hat{\boldsymbol{\beta}}$ is a least squares estimate of $\boldsymbol{\beta}$ iff

$$\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{M}\mathbf{y}.$$

Consequently the fitted values $\hat{\mathbf{Y}} = \mathbf{M}\mathbf{y}$ and residuals $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\mathbf{y}$ are the same for every least squares estimate.

Proof. Apply the projection theorem (Theorem 6.3.1) with $\mathcal{S} = \mathcal{C}(\mathbf{X})$ and $\mathbf{u} = \mathbf{X}\mathbf{b}$. By Eq. 6.3.1,

$$\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \|\mathbf{y} - \mathbf{M}\mathbf{y}\|^2 + \|\mathbf{M}\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 \quad \text{for every } \mathbf{b} \in \mathbb{R}^p.$$

The first term does not involve $\mathbf{b}$, and the second is zero iff $\mathbf{X}\mathbf{b} = \mathbf{M}\mathbf{y}$. Such a $\mathbf{b}$ exists because $\mathbf{M}\mathbf{y} \in \mathcal{C}(\mathbf{X})$. ■

The orthogonality half of the projection theorem turns this into linear equations. The residual $\mathbf{y} - \mathbf{X}\mathbf{b}$ is orthogonal to $\mathcal{C}(\mathbf{X})$ iff it is orthogonal to every column of $\mathbf{X}$, that is, iff $\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\mathbf{b}) = \mathbf{0}$.

Theorem 6.4.2. Normal equations

The **normal equations**

$$\mathbf{X}^\top \mathbf{X}\mathbf{b} = \mathbf{X}^\top \mathbf{y} \tag{6.4.1}$$

are consistent for every $\mathbf{y}$, and their solutions are exactly the least squares estimates. One solution is $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^- \mathbf{X}^\top \mathbf{y}$, for any generalized inverse, and the set of all solutions is

$$\hat{\boldsymbol{\beta}} + \mathcal{N}(\mathbf{X}) = \{\hat{\boldsymbol{\beta}} + \mathbf{v} : \mathbf{X}\mathbf{v} = \mathbf{0}\}.$$

In particular, the least squares estimate is unique iff $\text{rank}(\mathbf{X}) = p$, and then $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$.

Proof. A vector $\mathbf{b}$ solves Eq. 6.4.1 iff $\mathbf{y} - \mathbf{X}\mathbf{b} \perp \mathcal{C}(\mathbf{X})$. Since $\mathbf{X}\mathbf{b} \in \mathcal{C}(\mathbf{X})$ automatically, this holds iff $\mathbf{y} = \mathbf{X}\mathbf{b} + (\mathbf{y} - \mathbf{X}\mathbf{b})$ is the orthogonal decomposition of $\mathbf{y}$ (Theorem 6.2.3), that is, iff $\mathbf{X}\mathbf{b} = \mathbf{M}\mathbf{y}$. By Theorem 6.4.1 this is the same as $\mathbf{b}$ being a least squares estimate. For

the particular solution, $\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} = \mathbf{M}\mathbf{y}$ by Theorem 6.3.6. Finally, $\mathbf{X}\mathbf{b} = \mathbf{M}\mathbf{y} = \mathbf{X}\hat{\boldsymbol{\beta}}$ iff $\mathbf{X}(\mathbf{b} - \hat{\boldsymbol{\beta}}) = \mathbf{0}$, and $\mathcal{N}(\mathbf{X}) = \{\mathbf{0}\}$ iff $\text{rank}(\mathbf{X}) = p$ by Eq. 6.2.1. ■

The logic of this proof runs opposite to the usual calculus derivation, and the difference matters. Setting a gradient to zero finds stationary points. One then has to argue separately that they are minima and that one exists. Here existence, minimality and the full solution set all come from the decomposition $\mathbb{R}^n = \mathcal{C}(\mathbf{X}) \oplus \mathcal{C}(\mathbf{X})^\perp$. The consistency of the normal equations, which can look surprising when $\mathbf{X}^\top \mathbf{X}$ is singular, is just the statement $\mathbf{X}^\top \mathbf{y} \in \mathcal{C}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X}^\top \mathbf{X})$ from Corollary 6.2.4.

6.4.2 Fitted values and residuals

Collecting the consequences for $\hat{\mathbf{Y}} = \mathbf{M}\mathbf{y}$ and $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\mathbf{y}$:

Proposition 6.4.3

For any data vector $\mathbf{y}$:

- (a) $\mathbf{X}^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{0}$ and $\hat{\mathbf{Y}}^\top \hat{\boldsymbol{\varepsilon}} = 0$;
- (b) if $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ (for example, the model has an intercept), then $\sum_i \hat{\varepsilon}_i = 0$ and the mean of the fitted values is $\bar{y}$;
- (c) $\|\mathbf{y}\|^2 = \|\hat{\mathbf{Y}}\|^2 + \|\hat{\boldsymbol{\varepsilon}}\|^2$;
- (d) $\|\hat{\boldsymbol{\varepsilon}}\|^2 = \mathbf{y}^\top (\mathbf{I} - \mathbf{M})\mathbf{y}$ and $\|\hat{\mathbf{Y}}\|^2 = \mathbf{y}^\top \mathbf{M}\mathbf{y}$;
- (e) if $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ for some $\boldsymbol{\beta}$, then $\hat{\mathbf{Y}} = \mathbf{X}\boldsymbol{\beta} + \mathbf{M}\boldsymbol{\varepsilon}$ and $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon}$.

Proof.

- (a) is the orthogonality of $\hat{\boldsymbol{\varepsilon}}$ to $\mathcal{C}(\mathbf{X})$, and $\hat{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$. For (b), $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ gives $\mathbf{1}^\top \hat{\boldsymbol{\varepsilon}} = 0$, and then $\mathbf{1}^\top \hat{\mathbf{Y}} = \mathbf{1}^\top \mathbf{y}$. (c) is Pythagoras. (d) uses symmetry and idempotence: $\|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2 = \mathbf{y}^\top (\mathbf{I} - \mathbf{M})^\top (\mathbf{I} - \mathbf{M})\mathbf{y} = \mathbf{y}^\top (\mathbf{I} - \mathbf{M})\mathbf{y}$. For (e), $\mathbf{M}\mathbf{X} = \mathbf{X}$ and $(\mathbf{I} - \mathbf{M})\mathbf{X} = \mathbf{0}$. ■

Part (e) is simple but important. The residual vector does not depend on $\boldsymbol{\beta}$ at all. It is the projection of the unobservable error vector onto the $(n-r)$ -dimensional space $\mathcal{C}(\mathbf{X})^\perp$. Everything about σ^2 that the data can reveal must therefore come from $(\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon}$. If $E(\boldsymbol{\varepsilon}) = \mathbf{0}$ and $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$, the expectation of a quadratic form (Chapter 2) gives

$$E\|\hat{\boldsymbol{\varepsilon}}\|^2 = E[\boldsymbol{\varepsilon}^\top (\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon}] = \sigma^2 \text{tr}(\mathbf{I} - \mathbf{M}) = \sigma^2(n - r),$$

using Proposition 6.3.4. So $\|\hat{\boldsymbol{\varepsilon}}\|^2/(n - r)$ is unbiased for σ^2 . The divisor is the dimension of the space in which the residuals live, not the number of observations and not $n - p$. Chapter 7 develops the sampling theory, and Chapter 9 develops the decomposition.

6.4.3 Coefficients are not unique; fitted values are

When $\text{rank}(\mathbf{X}) < p$, Theorem 6.4.2 gives a whole affine subspace of least squares estimates, one for each generalized inverse and more besides. Statistical software has to pick one. Some

programs drop columns that are linearly dependent on earlier ones. Others impose “side conditions” such as a zero coefficient for a reference level or effects that sum to zero. Others report the minimum-norm solution. These choices look very different in the printed coefficient table. The geometry says they are all describing the same fitted vector.

Example. Four answers, one fit

Nine observations fall in three groups of sizes 3, 2, 4 with group means 5.700, 7.800 and 4.300. The model matrix $\mathbf{X} = [\mathbf{1}, \mathbf{z}_1, \mathbf{z}_2, \mathbf{z}_3]$ has $p = 4$ columns and rank 3, so $\mathbf{X}^\top \mathbf{X}$ is singular. We compute $\hat{\boldsymbol{\beta}} = \mathbf{G}\mathbf{X}^\top \mathbf{y}$ for four generalized inverses $\mathbf{G}$ of $\mathbf{X}^\top \mathbf{X}$: the Moore–Penrose inverse; the inverse obtained by deleting the intercept row and column (Theorem 1.3.2); the one obtained by deleting the first group’s row and column; and a deliberately strange, non-symmetric one built from the general formula of Exercise (B2).

Generalized inverse	$\hat{\beta}_0$	$\hat{\beta}_1$	$\hat{\beta}_2$	$\hat{\beta}_3$	$\hat{\beta}_1 - \hat{\beta}_2$	$\max \mathbf{M} - \mathbf{M}_{\text{MP}} $
Moore– Penrose	4.450	1.250	3.350	-0.150	-2.100	0
drop intercept	0.000	5.700	7.800	4.300	-2.100	1.1×10^{-16}
drop group 1	5.700	0.000	2.100	-1.400	-2.100	1.1×10^{-16}
random	-9.102	14.802	16.902	13.402	-2.100	1.8×10^{-15}

The coefficient vectors disagree wildly, including in sign. Yet each produces the same projection matrix up to rounding error, and hence the same fitted values (the group means) and residuals. The difference $\hat{\beta}_1 - \hat{\beta}_2$ is the same in every row, and so is $\hat{\beta}_0 + \hat{\beta}_k$ for each k . Individual coefficients are not. The trace of $\mathbf{M}$ is 3.000, the rank of $\mathbf{X}$.

```
import numpy as np

groups = np.array([0, 0, 0, 1, 1, 2, 2, 2, 2]) # three groups, sizes 3, 2, 4
y = np.array([5.1, 6.3, 5.7, 8.2, 7.4, 4.0, 4.9, 3.8, 4.5])
n, g = len(y), 3
X = np.column_stack([np.ones(n), np.eye(g)[groups]]) # [1, indicator columns]
A = X.T @ X
print("n =", n, " p =", X.shape[1], " rank(X) =", np.linalg.matrix_rank(X))
```

[Run this code online](#)

```
def g_drop(j):
    """Generalized inverse obtained by deleting row and column j of A."""
    keep = [k for k in range(A.shape[0]) if k != j]
    G = np.zeros_like(A)
    G[np.ix_(keep, keep)] = np.linalg.inv(A[np.ix_(keep, keep)])
    return G

rng = np.random.default_rng(6)
G0 = np.linalg.pinv(A) # Moore-Penrose inverse
U, V = rng.normal(size=A.shape), rng.normal(size=A.shape)
G_random = G0 + (np.eye(4) - G0 @ A) @ U + V @ (np.eye(4) - A @ G0)
```

```
ginverses = {
    "Moore-Penrose": G0,
    "drop intercept": g_drop(0),
    "drop group 1": g_drop(1),
    "random": G_random,
}
```

[Run this code online](#)

```
M_ref = X @ G0 @ X.T
for name, G in ginverses.items():
    assert np.allclose(A @ G @ A, A)           # G really is a g-inverse
    beta = G @ X.T @ y
    M = X @ G @ X.T
    print(f"{name:15s} beta = {np.round(beta, 3)}",
          f" max|M - M_ref| = {np.abs(M - M_ref).max():.1e}",
          f" group1-group2 = {beta[1] - beta[2]:.3f}")
```

[Run this code online](#)

Which linear functions of $\hat{\beta}$ come out the same for every least squares estimate? The answer is a first look at estimability, the subject of [Chapter 8](#).

Theorem 6.4.4

For $\lambda \in \mathbb{R}^p$, the value $\lambda^\top \hat{\beta}$ is the same for every least squares estimate $\hat{\beta}$ (and every $\mathbf{y}$) iff $\lambda \in \mathcal{C}(\mathbf{X}^\top)$, that is, iff $\lambda^\top = \rho^\top \mathbf{X}$ for some $\rho \in \mathbb{R}^n$. In that case $\lambda^\top \hat{\beta} = \rho^\top \mathbf{M}\mathbf{y}$.

Proof. By [Theorem 6.4.2](#) the least squares estimates form $\hat{\beta} + \mathcal{N}(\mathbf{X})$. So $\lambda^\top \hat{\beta}$ is invariant iff $\lambda^\top \mathbf{v} = 0$ for all $\mathbf{v} \in \mathcal{N}(\mathbf{X})$, that is, iff $\lambda \in \mathcal{N}(\mathbf{X})^\perp = \mathcal{C}(\mathbf{X}^\top)$ ([Corollary 6.2.4](#)). If $\lambda^\top = \rho^\top \mathbf{X}$ then $\lambda^\top \hat{\beta} = \rho^\top \mathbf{X}\hat{\beta} = \rho^\top \mathbf{M}\mathbf{y}$. ■

In [Example \(Four answers, one fit\)](#), $\lambda = (0, 1, -1, 0)^\top$ is the difference between any row of $\mathbf{X}$ from group 1, $(1, 1, 0, 0)$, and any row from group 2, $(1, 0, 1, 0)$. So $\lambda \in \mathcal{C}(\mathbf{X}^\top)$ and $\hat{\beta}_1 - \hat{\beta}_2$ is invariant. For $\lambda = (0, 1, 0, 0)^\top$, note that $\rho^\top \mathbf{X}$ has first entry $\sum_i \rho_i$ and remaining entries equal to the sums of ρ_i within each group. Matching $(0, 1, 0, 0)$ would need the group sums to be 1, 0, 0 and their total to be 0, which is impossible. So $\hat{\beta}_1$ is not invariant.

Remark (Side conditions choose coordinates). Imposing $\beta_0 = 0$ or $\beta_1 = 0$ does not add information about the data. It picks one point from the affine set $\hat{\beta} + \mathcal{N}(\mathbf{X})$. A side condition $\mathbf{c}^\top \beta = 0$ picks exactly one point iff $\mathbf{c} \notin \mathcal{C}(\mathbf{X}^\top)$. A condition inside $\mathcal{C}(\mathbf{X}^\top)$ would constrain the fitted values and change the model rather than just its coordinates. [Theorem 8.4.2](#) makes this precise.

6.4.4 Exercises

6.4.4.1 A. Check your understanding

Exercise. A1

For Example (Three observations), write out the 3×3 matrix $\mathbf{M}$ with exact fractional entries, verify $\text{tr } \mathbf{M} = 2$, and recover the fitted values and residuals quoted in the text.

Exercise. A2

Show that the residuals sum to zero whenever $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, even if no column of $\mathbf{X}$ is constant. Give an example of a model without such a column in which this happens, and one in which the residuals do not sum to zero.

6.4.4.2 B. Practice

Exercise. B1

Suppose $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ and $\text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I}$. Show that $E(\hat{\mathbf{Y}}) = \mathbf{X}\boldsymbol{\beta}$, $\text{Cov}(\hat{\mathbf{Y}}) = \sigma^2\mathbf{M}$, $\text{Cov}(\hat{\boldsymbol{\varepsilon}}) = \sigma^2(\mathbf{I} - \mathbf{M})$ and $\text{Cov}(\hat{\mathbf{Y}}, \hat{\boldsymbol{\varepsilon}}) = \mathbf{0}$. Which of these statements survive when $\text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{V}$ for a general positive definite $\mathbf{V}$?

Exercise. B2

Let $\text{rank}(\mathbf{X}) = r < p$ and let $\mathbf{C}$ be a $(p - r) \times p$ matrix. Show that the constrained problem “minimize $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|$ subject to $\mathbf{C}\mathbf{b} = \mathbf{0}$ ” has a unique solution for every $\mathbf{y}$, and that its fitted values are $\mathbf{M}\mathbf{y}$, iff $\text{rank}(\begin{smallmatrix} \mathbf{X} \\ \mathbf{C} \end{smallmatrix}) = p$. Show that the condition implies $\mathcal{C}(\mathbf{C}^\top) \cap \mathcal{C}(\mathbf{X}^\top) = \{\mathbf{0}\}$, and interpret: the constraints must not restrict any estimable function.

6.4.4.3 C. Going deeper

Exercise. C1

Suppose $\mathbf{X}^\top \mathbf{y} \neq \mathbf{0}$. Show that every least squares estimate $\mathbf{b}$ can be written as $\mathbf{G}\mathbf{X}^\top \mathbf{y}$ for some generalized inverse $\mathbf{G}$ of $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$. *Hint:* with $\mathbf{c} = \mathbf{X}^\top \mathbf{y}$, try $\mathbf{G} = \mathbf{G}_0 + (\mathbf{b} - \mathbf{G}_0\mathbf{c})\mathbf{c}^\top / (\mathbf{c}^\top \mathbf{c})$. What goes wrong when $\mathbf{X}^\top \mathbf{y} = \mathbf{0}$?